

Jiaxin Zhang

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Profile

Senior AI researcher at Tencent London with a PhD in Computer and Information Sciences. Develops LLM post-training and agent systems for real-time game interaction, with experience in synthetic data, reasoning distillation, tool calling and multi-agent execution. Research spans mathematical reasoning, multimodal game agents and temporal memory. First-author publications at NeurIPS, ACL Findings and NAACL Findings; internship mentorship leading to EMNLP 2026 Oral and SIGGRAPH 2025 papers. UK Global Talent visa holder.

Work Experience

Tencent IEG, LIGHTSPEED STUDIOS

Aug 2024 - Present

Senior NLP Researcher (T9) | London, UK

- Lead applied LLM and agent research for Game for Peace, spanning instruction understanding, Qwen 4B fine-tuning, reasoning distillation and structured action generation.
- Develop synthetic-data and evaluation workflows for implicit instructions, spatial references, multi-turn context and compound tasks; translate research into real-time game-agent capabilities.
- Mentored research interns on two published or accepted projects: Time is Not a Label (EMNLP 2026 Main Conference, Oral; corresponding author) and CueTip (SIGGRAPH 2025). Provide guidance on model development, experiments and code review.

University of Strathclyde

Feb 2022 - Oct 2024

Teaching Assistant, CS985 Machine Learning for Data Analytics | Glasgow, UK

- Led postgraduate machine-learning labs and coordinated teaching assistants on assignment and exam marking.

TCL Technology

May 2019 - Sep 2020

NLP Researcher | China

- Built a conversational dataset of 60,000+ multi-turn dialogues and 1M+ utterances; automated data preparation and training, reducing manual preprocessing time by over 70%.
- Developed BERT retrieval (+24.7% F1), Bucket-Search (response time below 20 ms) and dialogue components integrated into TCL smart TVs.

Selected Projects

Auto-gaming: Visual Perception and Closed-loop Execution

Ongoing research

- Developed a vision-based FPS agent prototype that connects target localization and state assessment with reusable action skills for closed-loop task execution.
- Exploring automatic refinement of visual prompts and state hypotheses from failed interactions, aiming to reduce manual tuning and improve robustness across objects, maps and tasks.

Game for Peace: LLM Instruction Following and Multi-agent Control

Tencent

- Built synthetic-data pipelines covering hierarchical intent labels, implicit long-tail commands, spatial references, negative examples and compound task stacks.
- Distilled tactical reasoning from larger LLMs into Qwen 4B and fine-tuned tool calling and structured action outputs to ground ambiguous player instructions in executable game APIs.
- Engineered an asyncio-based planner and task-stack runtime for sequential, conditional and parallel commands, including speaker-aware context handling for two-player, two-AI coordination.

Selected Publications

Google Scholar: 627 citations; h-index 7 (29 September 2026).

1. **Time is Not a Label: Continuous Phase Rotation for Temporal Knowledge Graphs and Agentic Memory**
Weixian Waylon Li, **Jiaxin Zhang**, Xianan Jim Yang, Tiejun Ma, Yiwon Guo.
[EMNLP 2026 Main Conference - Oral](#) | [Corresponding author & research intern mentor](#)
RoMem models temporal evolution through continuous phase rotation and relation-dependent semantic speed.
2. **From System 1 to System 2: A Survey of Reasoning Large Language Models**
Duzhen Zhang, Zhong-Zhi Li, Ming-Liang Zhang, **Jiaxin Zhang**, et al.
[IEEE TPAMI, 48\(3\):3335-3354, 2026 \(online 2025\)](#)
3. **CueTip: An Interactive and Explainable Physics-aware Pool Assistant**
Sean Memery, Kevin Denamganāi, **Jiaxin Zhang**, Zehai Tu, Yiwon Guo, Kartic Subr.
[ACM SIGGRAPH 2025 Conference Papers](#) | [Research intern mentor](#)
Combines language-model interaction, physics simulation and expert rules for grounded shot advice and explanations.
4. **CMMaTH: A Chinese Multi-modal Math Skill Evaluation Benchmark for Foundation Models**
Zhongzhi Li, Ming-Liang Zhang, ..., **Jiaxin Zhang**, Cheng-Lin Liu.
[COLING 2025](#)
5. **GeoEval: Benchmark for Evaluating LLMs and Multi-Modal Models on Geometry Problem-Solving**
Jiaxin Zhang, Zhong-Zhi Li, Ming-Liang Zhang, Fei Yin, Cheng-Lin Liu, Yashar Moshfeghi.
[Findings of ACL 2024](#) | [First author](#)
6. **GOLD: Geometry Problem Solver with Natural Language Description**
Jiaxin Zhang, Yashar Moshfeghi.
[Findings of NAACL 2024](#) | [First author](#)
7. **ELASTIC: Numerical Reasoning with Adaptive Symbolic Compiler**
Jiaxin Zhang, Yashar Moshfeghi.
[NeurIPS 2022](#) | [First author](#)

Education

University of Strathclyde <i>PhD in Computer and Information Sciences</i> Thesis: Advancing Mathematical Reasoning with Deep Learning: From Numerical Insights to Geometrical Understanding. Nominated for the SICSA 2025 Best Dissertation Award.	Oct 2020 - Oct 2024
University of Sheffield <i>MSc in Advanced Computer Science</i>	Sep 2017 - Oct 2018
Chang'an University <i>BSc in Network Engineering, Computer Science</i>	Sep 2011 - Jul 2015

Honors & Awards

NeurIPS 2022 Scholar Award – Conference registration and accommodation support.
SCDT Research Studentship – Fully funded 3.5-year PhD studentship, 2020–2024.

Skills

LLMs & agents: Supervised fine-tuning, reasoning distillation, synthetic data, tool calling, spatial grounding, multi-agent orchestration, evaluation.
Engineering: Python, SQL, PyTorch, Transformers, Accelerate, DeepSpeed, LLaMA-Factory, scikit-learn, asyncio, Git, Bash, LaTeX.
Languages: Chinese (native); English (proficient).

Academic Service

Reviewer for NeurIPS and EMNLP.